



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Strategic insights from Python, SQL & Power BI

Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating only

\$59.76

Avg Purchase

Features span demographics, purchase details, and shopping behavior — including subscription status, discounts, and review ratings.

Data Preparation in Python

01

Load & Explore

Imported via pandas; `df.info()` and `describe()`

02

Clean & Standardize

Imputed Review Rating by category median; renamed to `snake_case`

03

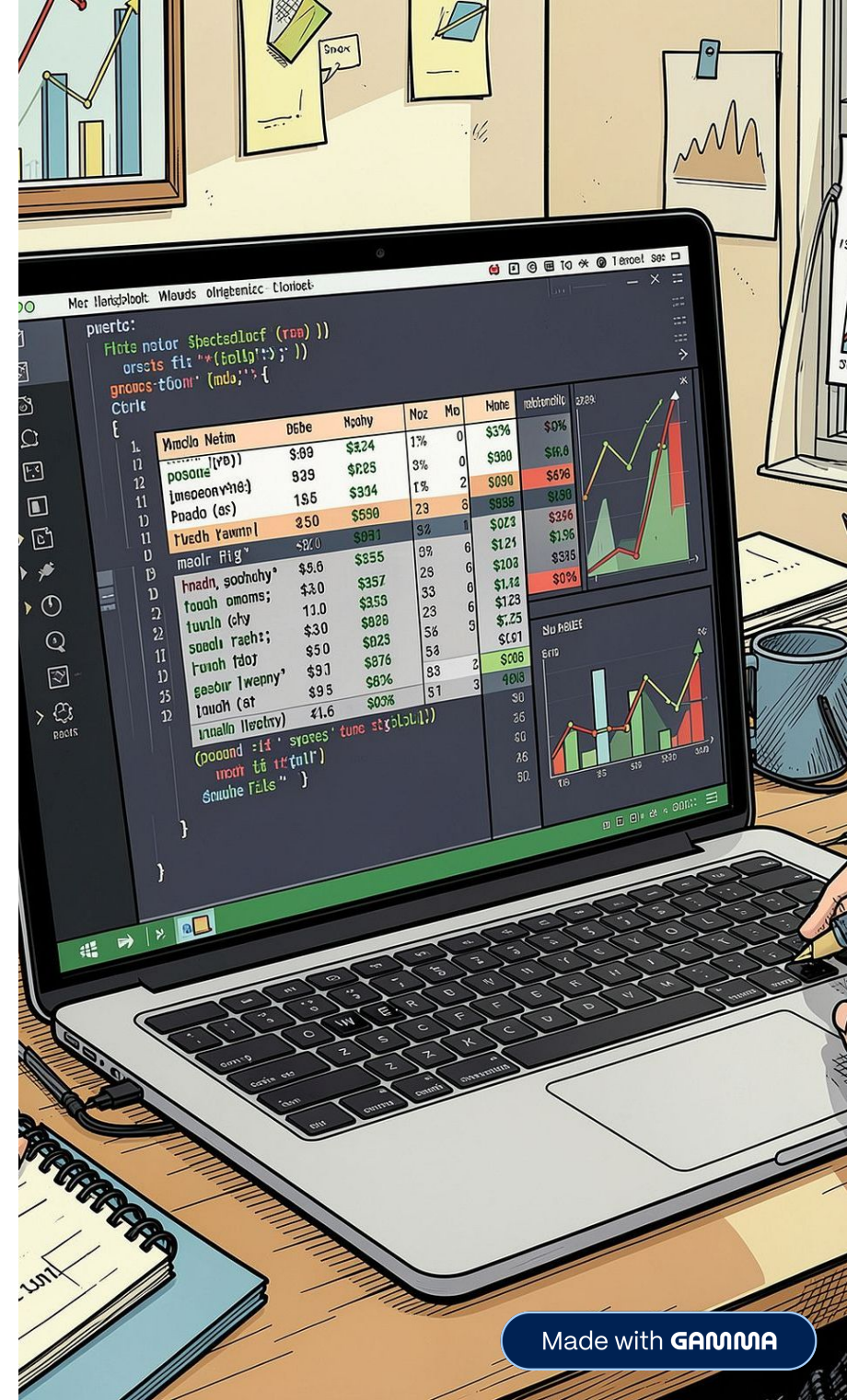
Feature Engineer

Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`

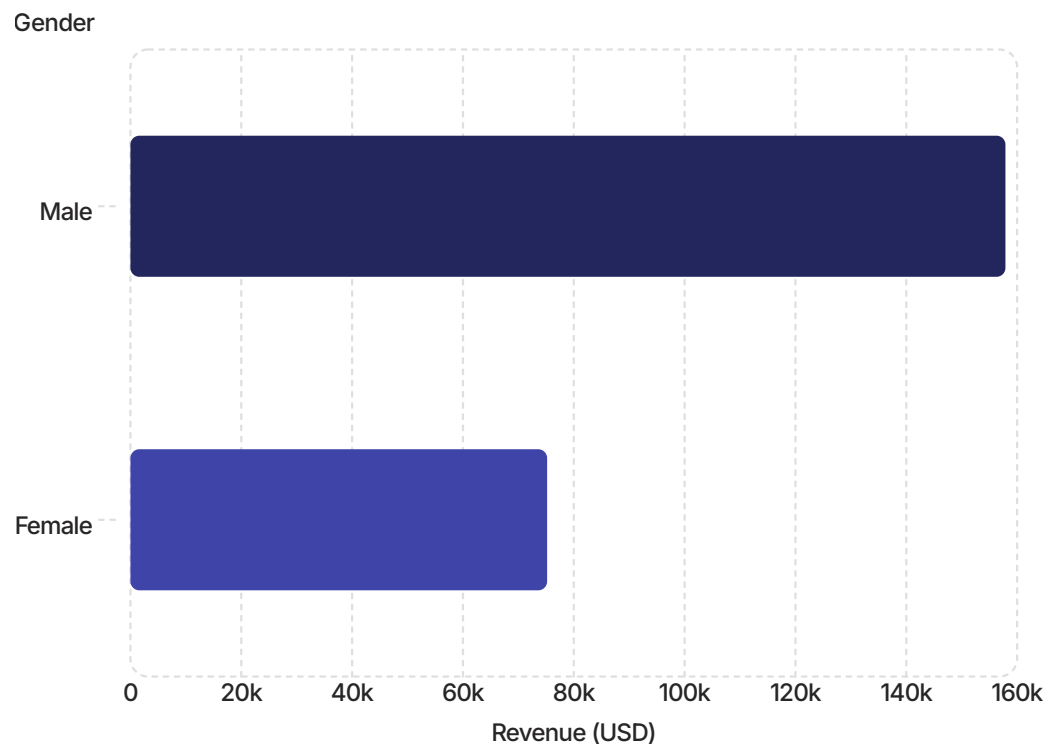
04

Load to DB

Cleaned DataFrame loaded into PostgreSQL for SQL analysis



Revenue by Gender



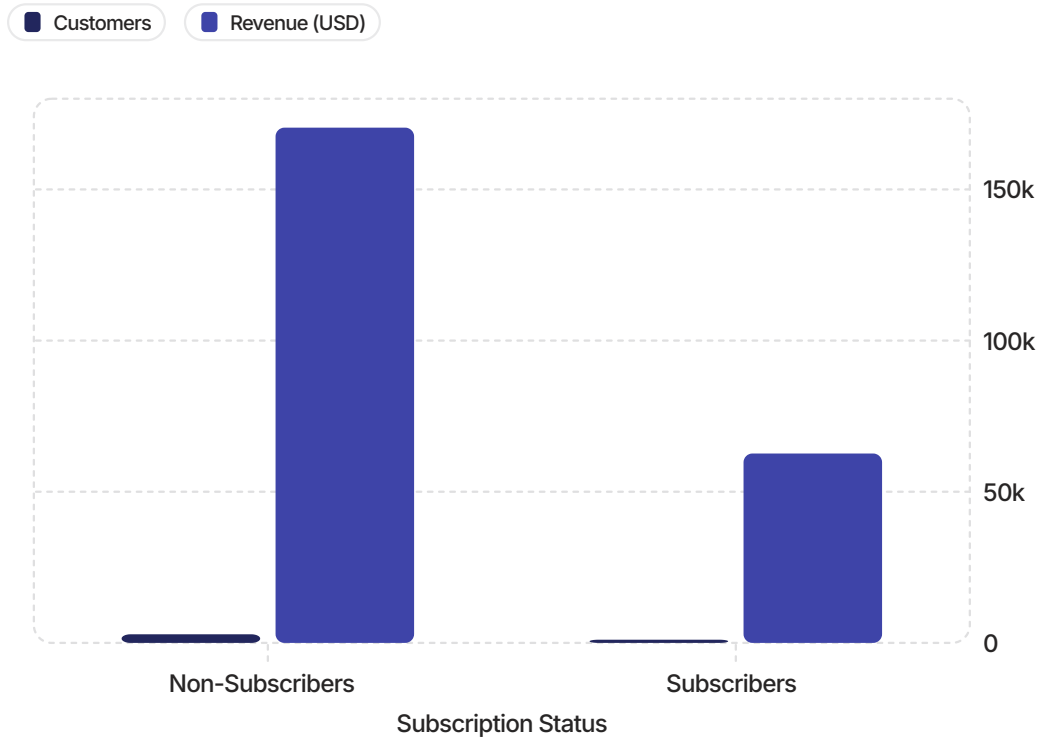
Male customers drive 2× more revenue

Male: **\$157,890**

Female: **\$75,191**

Male customers also represent 68% of all transactions (2,652 of 3,900).

Subscribers vs. Non-Subscribers



Key Insight

Non-subscribers outnumber subscribers **3:1** and generate **73%** of total revenue.

Average spend is nearly identical: \$59.87 vs \$59.49 — subscription doesn't yet drive higher spend.

Top Products & Categories

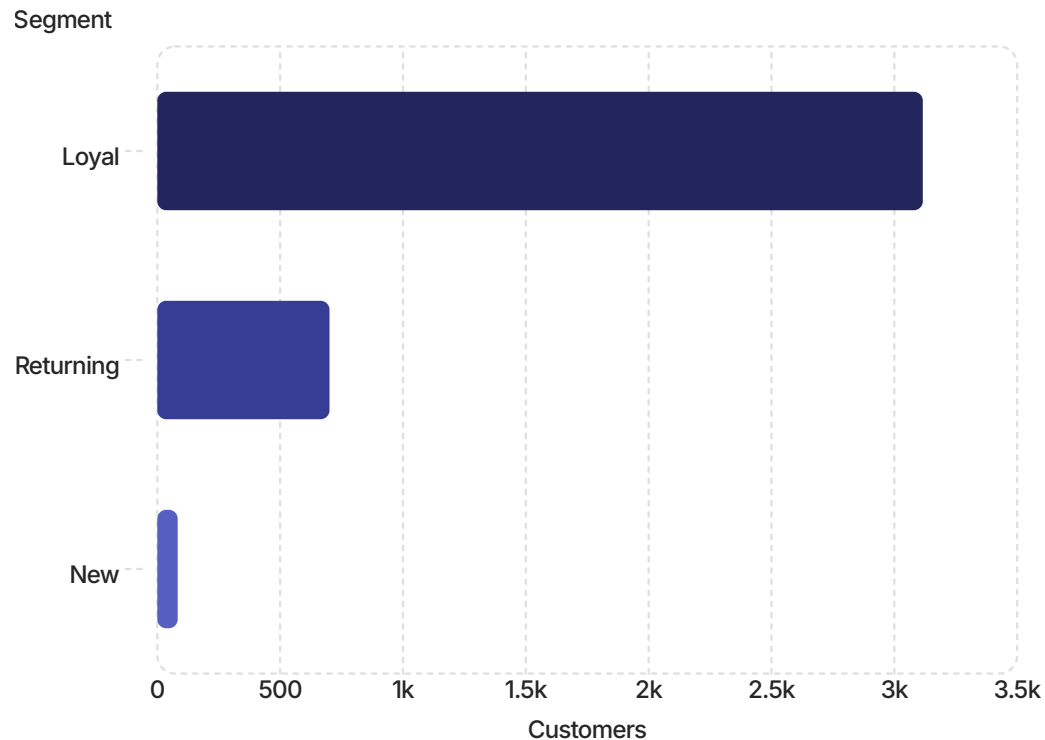
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Purchased by Category

- **Accessories:** Jewelry (171), Sunglasses (161)
- **Clothing:** Blouse (171), Pants (171)
- **Footwear:** Sandals (160), Shoes (150)
- **Outerwear:** Jacket (163), Coat (161)

Customer Segmentation

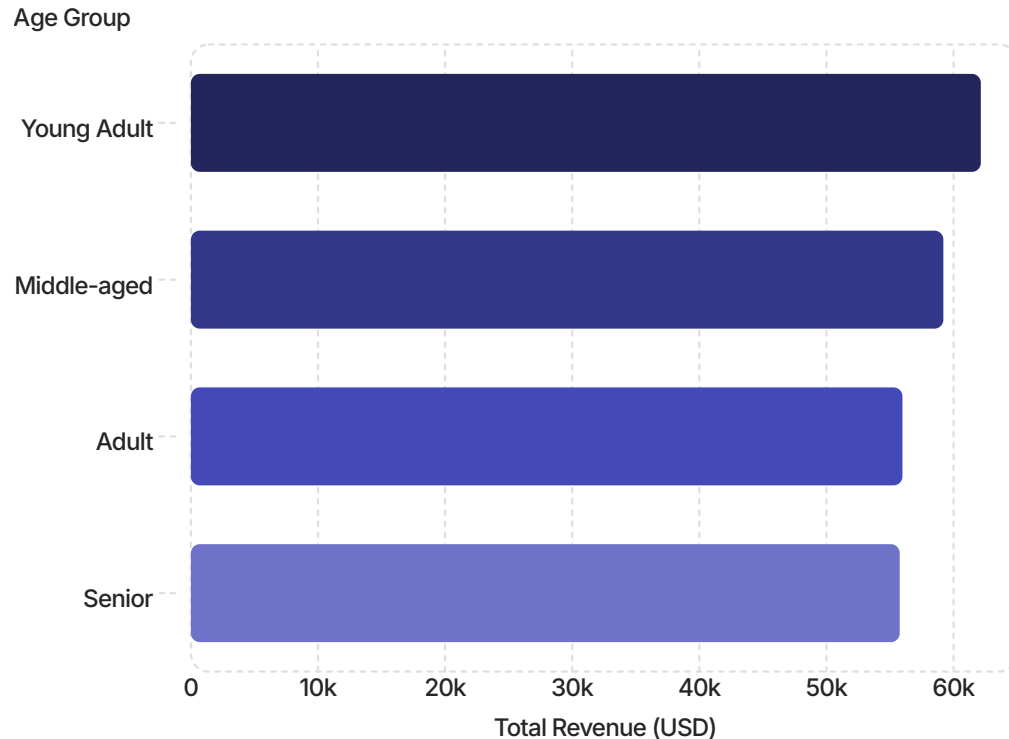


80% are Loyal Customers

Loyal: **3,116** · Returning: **701** · New: **83**

Of customers with 5+ purchases, 958 are subscribers vs. 2,518 non-subscribers — repeat buyers aren't yet converting to subscriptions.

Revenue by Age Group



Young Adults Lead

Young Adults contribute the highest revenue at **\$62,143**, closely followed by Middle-aged at **\$59,197**.

Revenue is relatively evenly distributed — all four age groups contribute within ~\$6K of each other.



Power BI Dashboard

Key Metrics

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Revenue by Category

Clothing \$100K · Accessories
\$75K · Footwear \$30K ·
Outerwear \$15K

Subscription Split

27% subscribers vs 73% non-subscribers

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

